Ghost Racer — Proposed Application States

COMP 7082 Project · 12 October 2026 · Based on the repository at commit 52fec81 (branch Ethan)

This document proposes the set of states for the Ghost Racer app, for use in the project's state diagram. The states are grouped into the main user journey, a GPS region that runs in parallel while recording or racing, and the lifecycle of stored data.

The last column of each table records where the state stands in the code today: **Yes** means it is implemented, **Partial** means it is present but incomplete, and **New** means it is proposed and not yet in the code.

1. Primary states (user journey)

#	State	Entered when	In code today
1	**Launching**	App opens and the saved session is being restored	**Yes**: `loading`
2	**Browsing as guest**	No session	**Yes**
3	**Browsing signed in**	Session present	**Yes**
4	**Sign-in required**	Guest taps a gated action (Race, Save & Challenge)	**Yes**: `prompt`
5	**Authenticating**	Sign-in or sign-up form submitted	**Yes**: `busy`
6	**Awaiting email confirmation**	Sign-up succeeded but is unconfirmed	**Partial**: a message only
7	**Ready to record**	Record tab open, activity (run or bike) chosen	**Yes**: `idle`
8	**Recording**	Start pressed	**Yes**: `recording`
9	**Recording paused**	Pause pressed	**Yes**: `paused`
10	**Reviewing draft**	Finish pressed; user can name, save or discard	**Yes**: `draft`
11	**Race ready**	Route chosen from Home	**Yes**
12	**Racing**	Start Race pressed	**Partial**: no ghost comparison shown
13	**Race paused**	Pause pressed during a race	**Yes**
14	**Race finished**	Runner reaches the end of the route	**New**
15	**Race abandoned**	End Race pressed before the finish	**Partial**: nothing is saved

Sub-states of Racing (state 12)

Sub-state	Meaning	In code today
Ahead	Runner reaches the nearest ghost point in less time than the ghost did	**Partial**: computed, not displayed
Behind	Runner reaches the nearest ghost point in more time than the ghost did	**Partial**: computed, not displayed
Off route	Runner is farther than a set distance from the route	**New**

The ahead/behind value is already computed by `computeGhostDelta` in `src/services/ghostCompare.ts`.

2. GPS states (parallel with states 7 to 13)

State	Trigger	In code today
Requesting permission	First use of location	**Partial**: declared, not wired to any screen
Permission denied	User declines	**Partial**: recorder silently drops back to idle
Acquiring fix	Permission granted, no fix yet	**New**
Live	Fix received	**Partial**: declared, not wired to any screen
Weak signal	Accuracy worse than 25 m	**Partial**: `flagAccuracy` exists but is not shown
Signal lost	No fix for 5 seconds	**Partial**: declared as `stale`, never set
Error	Location service fails	**Partial**: declared, not wired to any screen

The declared states come from `TrackingStatus` in `src/types/gps.ts`. The hook that uses it (`useGpsTracking`) is not mounted by any screen.

3. Data lifecycle states

Route

State	Meaning	In code today
Draft	Recorded but not yet saved; held in memory	**Yes**: `RouteDraft`
Saved locally	Stored on the device	**Yes**
Published (active)	Uploaded and visible to other users	**Partial**: `routes.status` defaults to `active`; no upload path
Archived	Hidden from discovery without deleting its runs	**New**

Run

State	Meaning	In code today
In progress	A race is under way	**Partial**: exists only in memory
Completed	Race finished and stored	**Partial**: `runs` table exists; app does not save runs
Record	Fastest run on its route	**Partial**: `runs.is_record` column only
Flagged	Suspicious run excluded from the leaderboard	**Partial**: `runs.flagged` column only

Schema references are to `db/schema.sql`. The “Public” filter chip on the Home screen is not yet connected to anything.

4. Gaps in the current code that these states expose

- **Race has no ghost.** `RaceScreen.tsx` drives the route recorder instead of `useGhostRace`, so states 12, 14 and 15 have no ahead/behind readout and no result.
- **Recording and racing are not exclusive.** Both screens share one recorder created in `AppShell.tsx`. From reading the code (not yet reproduced on a device), opening a race in the middle of a recording would show that recording as the race, and End Race would discard it.
- **Permission denial is invisible.** The recorder shows Recording before the permission request resolves, then returns to idle with only a console warning.